Program 8: Create Kubernetes Pods using YAML Descriptors

Aim

To create a Kubernetes **Pod** using **YAML descriptors** and deploy a simple **Node.js application** inside the Pod using a **ConfigMap**.

Prerequisites

- Minikube installed and running
- kubectl configured
- Basic knowledge of YAML

Step 1: Start Minikube

minikube start

Step 2: Create Node.js Application (server.js)

```
const http = require("http");

const port = 3000;



const server = http.createServer((req, res) => {

  res.end("Hello from Node.js running inside Kubernetes!");

});
```

server.listen(port, () => {

  console.log(`Server running at port ${port}`);

});

Explanation

- Creates a simple HTTP server
- Listens on port 3000
- Returns a response when accessed

Step 3: Create ConfigMap YAML (nodejs-configmap.yaml)

apiVersion: v1

kind: ConfigMap

metadata:

  name: nodejs-app-config

data:

```
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Explanation

- ConfigMap stores non-sensitive configuration data
- server.js is stored inside the ConfigMap
- Allows code to be injected into the Pod without building a Docker image

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Step 4: Create Pod YAML (nodejs-pod.yaml)

```
apiVersion: v1
kind: Pod
metadata:
  name: program-8
spec:
  containers:
  - name: nodejs
    image: node:18
    command: ["node", "/usr/src/app/server.js"]
    volumeMounts:
    - name: app-code
      mountPath: /usr/src/app
  volumes:
  - name: app-code
    configMap:
```

name: nodejs-app-config

pods.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: my-pod
spec:
  containers:
    - name: node
      image: node:14-alpine
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: config-volume
          mountPath: /usr/src/app
  volumes:
    - name: config-volume
      configMap:
        name: configmap
```

Explanation

- Pod is the smallest deployable unit in Kubernetes
- Node.js container runs the application
- ConfigMap is mounted as a volume inside the container

Step 5: Apply the YAML Files

kubectl apply -f nodejs-configmap.yaml

kubectl apply -f nodejs-pod.yaml

Step 6: Verify Pod Status

kubectl get pods

Expected status: `Running`

Step 7: Access the Application

kubectl port-forward pod/program-8 3000:3000

Open browser and visit:

http://localhost:3000

Expected Output

Hello from Node.js running inside Kubernetes!

Conclusion

This experiment demonstrates how to create a Kubernetes Pod using YAML descriptors, use